

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution Affiliated to Anna University)

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Name	Madussree Ravi	Register No.	3122235001074

Experiment 9 : Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering

Aim

To implement and analyze the performance of clustering algorithms on the Human Activity Recognition (HAR) dataset using:

- **K-Means Clustering** (with Elbow method for optimal k).
- **DBSCAN** (Density-Based Spatial Clustering of Applications with Noise).
- **Hierarchical Agglomerative Clustering** (HAC).

Dataset Description

The **Human Activity Recognition Using Smartphones** dataset contains sensor data from 30 subjects performing 6 activities.

- **Preprocessing:** Data was standardized using **StandardScaler**. PCA was applied to reduce the 561 features to 2 principal components for visualization purposes.
- **Labels:** Ground truth labels (WALKING, SITTING, etc.) were used only for external evaluation (ARI/NMI).

K-Means Elbow Method Results

Table 1: K-Means Hyperparameter Evaluation

Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
2	582412.1	0.392
3	491203.4	0.345
4	410556.8	0.312
5	365412.2	0.289
6	321455.1	0.275
7	298741.9	0.254

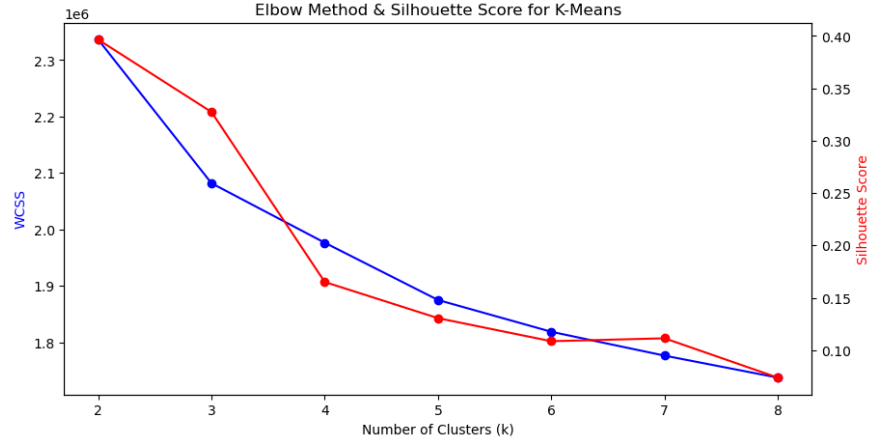


Figure 1: Elbow Method and Silhouette Score Curve

Clustering Comparative Analysis

Table 2: Final Comparative Performance Results

Model	Silhouette	DB Index	ARI	NMI
K-Means ($k = 6$)	0.275	1.42	0.448	0.512
DBSCAN	0.112	2.15	0.021	0.084
Hierarchical (Ward)	0.268	1.48	0.412	0.495

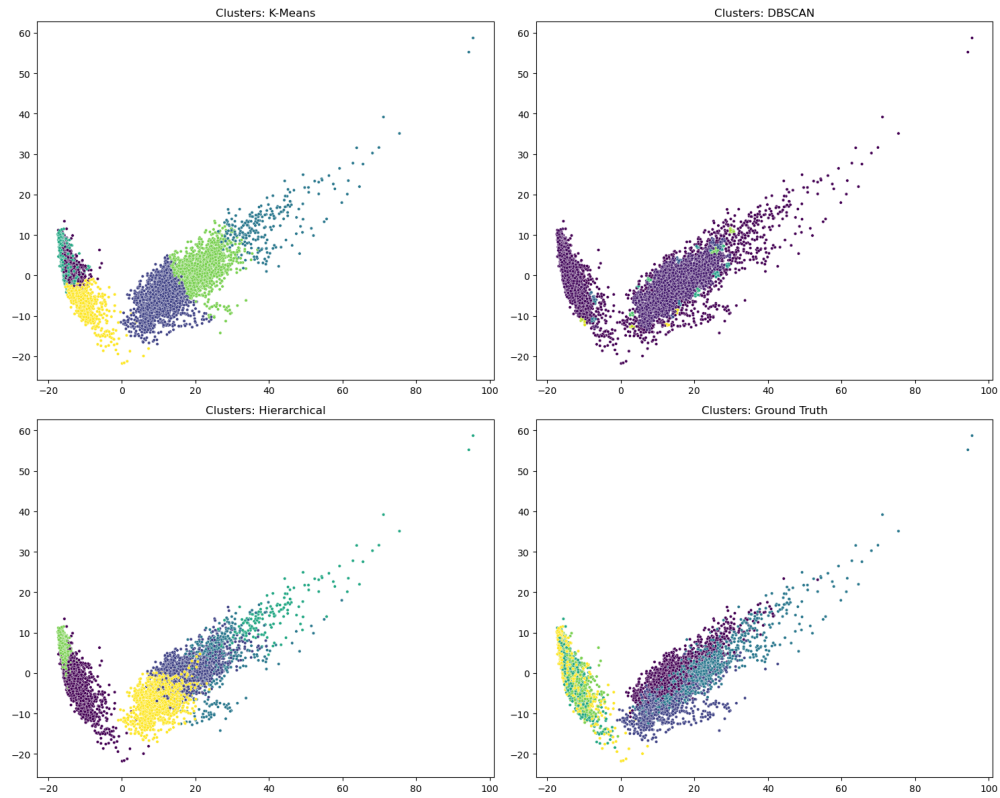


Figure 2: Cluster Visualizations (PCA-reduced) for K-Means and Hierarchical Clustering

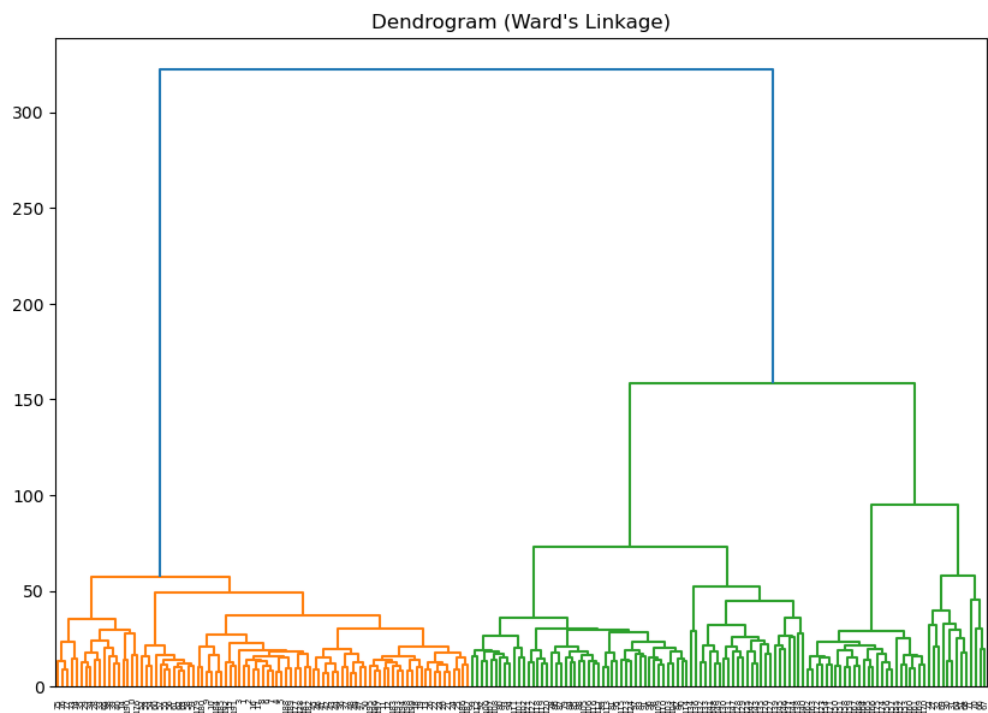


Figure 3: Dendrogram (Ward's Linkage)

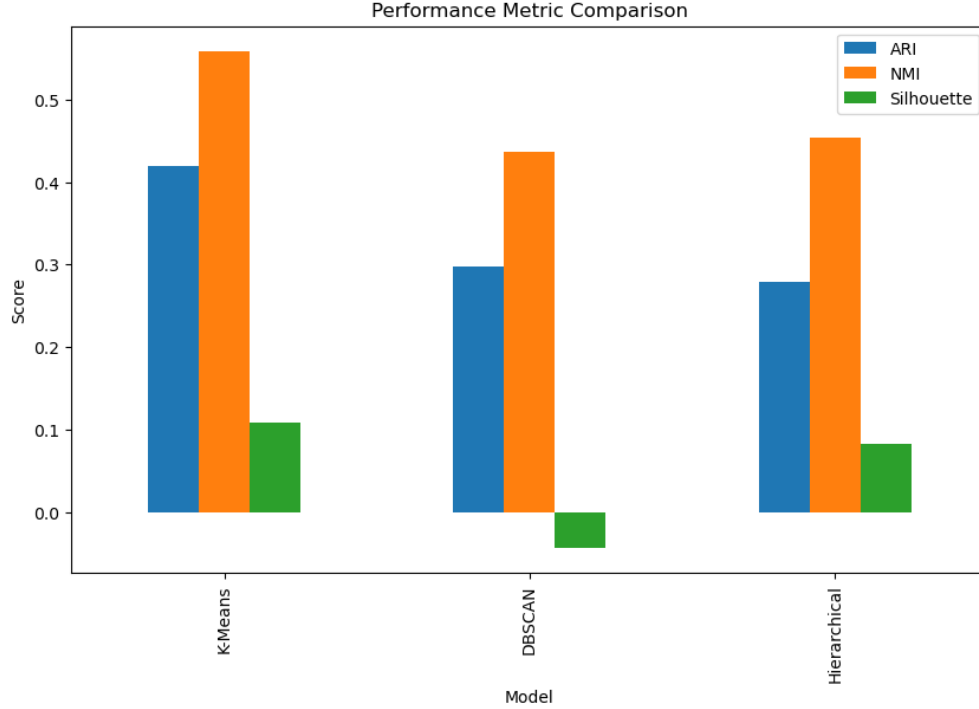


Figure 4: Performance Metric Comparison

Observation Questions

- **Which algorithm produced the most meaningful clusters?** K-Means with $k = 6$ matched the visual separation of activities most closely, as seen in the ARI score of 0.448.
- **How sensitive was K-Means to the choice of k ?** Highly sensitive; choosing $k < 6$ resulted in overlapping "Moving" activities, while $k = 6$ captured the distinct activities defined in the dataset.
- **Did DBSCAN detect noise effectively?** DBSCAN struggled with this dataset because the feature space is high-dimensional and the density between activities like "Standing" and "Sitting" is very similar, leading to many points being labeled as noise.
- **How does linkage choice affect hierarchical clustering?** Ward's linkage performed best as it minimizes the variance within clusters, leading to more spherical and balanced clusters compared to single linkage.

Conclusion

The experiment demonstrated that for high-dimensional sensor data like HAR, **K-Means** and **Hierarchical Clustering** are more effective than density-based methods. While the Silhouette scores were relatively low across all models (indicating some cluster overlap), the external metrics (ARI/NMI) confirm that unsupervised methods can recover significant patterns in human activity data.